Although writing a report is a skill you will learn in many other modules, this document will attempt to provide you with the expected structure for a Facilities Design report.

“Words(and pictures) sell, or they don’t”

A report is often the only thing you will have to present and sell your idea to either customer, colleague, or boss.

A Report is READ!!! So you must write so that your thinking logic is read, not assumed.

General tips:

Plan your report with a pen on some paper first, by figuring out what your Table of Contents (ToC) must/can be. The ToC must be **logically** structured – like telling a story, so ensure **coherence**.

While planning the ToC, separate and list and number different sections, Tables and Figures.

It will help if somebody *completely objective* reads the report and give feedback. If they cannot understand it, then neither will the examiner.

The structure:

Apart from the Cover page (with all relevant detail) and Appendices (declutter the report from too much detail)

1. Executive Summary

Must be concise yet comprehensive. No longer than one page in length. Brief overview of the current state (or Needs analysis), key design requirements, chosen design and motivation as well as implementation estimates.

2. Background

Provide context for your project. Distinguish between a new facility design or a current facility redesign, as this will influence your constraints.

3. Design Objectives

Explain and list current needs or problems with the current facility. Your main objective of the project will typically be: ‘To design a xxx(type) facility that (specify what it will be able to do xxx)’

Define the objectives (what is it that you are aiming to achieve) of the facility. [see Appendix for info on Objectives]

4. Design Requirement statements

See Appendix for info on Design Requirement statements. These have to do with capacity, process activities and storage policies and even space requirements. The better you can **quantify** the statements at this point the better you will be able to **evaluate** options to choose the best one and **measure** the design success at the end.

Make sure that you describe how you would know if your facility layout was good. If a facility does not meet the requirements, then it is not a good facility.

5. Assumptions and constraints

Explain assumptions you made or were given (if appropriate) as well as the constraints you are limited by. These can be space, budget or time limits. Also specific environmental, health, procedural, regulatory specifications etc. the facility needs to adhere to.

6. Detail design specifications

Give an overview of all the DATA that you researched, calculated, asked for or assumed. If you are redesigning a facility, specifying the current state is recommended to show improvement. You must collect data on everything! Sales volumes, demand, production capacity, process detail activities (this is where you need to show the Production process flow charts), interrelationships amongst activities, where is raw materials received and final products dispatched, how does material flow through the process, include storage areas specifics and policies, transport processes, specify equipment, labor etc. Also include all supporting processes eg. Power, water, offices, personnel, light, parking, amenities etc. See GETTING DATA GUIDANCE document also.

This is the heading where a lot of **calculations** are done, but you only place summaries here. It is under this heading that you need to be strategic about what you include in your report and what you place in an appendix. Tables work well especially if it shows calculations logic well.

Show **DETAIL workstation designs** as evidence that you understand the practical implications on that level of detail. These can then be used to support the bigger space requirements calculations especially for work AREAS or departments.

The goal of showing data and calculations is to provide a **logical traceable argument** for the decisions that you took when designing your facility. Your data must therefore be strategic to convince your reader that the thinking and analysis behind your final design is solid, rigorous, and trustworthy since it is based on researched data.

7. Design Alternatives

Give an overview of the alternatives that you considered and how you evaluated them. Include at least 3 alternatives on some aspect of your design.

Block layouts (indicate areas in blocks to check your dimensions relations) are very useful at this stage as you do not need a high degree of detail. Also indicate how you developed alternatives, based on **what logic**, criteria or method or approach (if possible). Basically – try and **lead the reader through your thinking process**. Be creative and make use of brainstorming techniques to develop original ideas.

8. Evaluating alternatives

Make use of **some** technique in doing an evaluation, so that it shows a systematic approach, not just liking some ideas better than others. Refer to the Objectives and Requirements to measure effectiveness of the design alternatives.

9. The final design

Present a Master Facility layout (a full conceptual layout plan of whole plot). This must logically flow from the Block layout chosen. Indicate scale (as line) and dimensions for the big areas such as Plot, buildings and outside parking. Use arrows to show material flow. Highlight blocks or areas using colours / shapes etc. You do not need fancy software but you do need to be creative.

10. Business case

In the business case for implementation the following should be included at least:

Capital cost estimates of buildings (based on cost/sq m) and equipment (as calculated). No need to include stocking the warehouses, Procurement functions will take care of that. Operational costs do not need to be included as a Facility Design project is usually a Capex project handed over to Operations when finished. The business case heading must also include Risks associated with implementation (and mitigation strategies) and a high level Schedule.

Below some tips on FORMATTING of your report

Formatting has a significant impact on the overall professionalism of your report.

- Use a standard referencing method.
- Please spell and grammar check your work. Don't rely on your own eye to pick up the mistakes – use the functionality in your Word processor or make use of the free version of Grammarly.
- 11 size font works well
- 1.5 line spacing works well
- Calibri, Arial, Times New Roman are some of the more traditional fonts
- Full justify your text – it makes it look much more professional
- Make strategic use of headings and sub-headings. Not too many and not too little.
- Avoid going more than 3 levels deep in sub-headings. If you are, restructure all your headings.
- Number Headings and sub-headings and check that it is correct when inserting the ToC
- Very often students make unnecessary use of Smart Art type graphics. Technical reports rarely have big, bold, colourful, blocked graphics like this.
- DO make use of bulleted lists, tables and figures where ever possible but construct them neatly.
- It is a good idea to always portray your ideas using figures and tables to make it easier for the reader to follow.
- Never include figures or tables without referring to them in the text. And ***always explain*** figures and tables, drawing the reader's attention to important details. The figures/tables do not replace the text.